CURBCOUNT · PROJECT PLAN

CurbCount project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/curbcount		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Opened in the kit 1.9.1 upgrade

CurbCount project plan

CurbCount is at stage P3 (TRL 3, paper): the design is constructable and fully documented, and the next gate (P4, TRL 4) is on hold under the portfolio cap. The decisions requested from Amish are the concept choice (O2) and the cold-site question (O1) in CBC-DEC-001. Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Problem and users	Design lead	CBC-PRB-001, CBC-URS-001	Done (v0.8; URS v0.1)
STEM check	Design lead	CBC-STM-001, `stem_check.py`	Done; S8 open for Amish
Concept options	Design lead	CBC-OPT-001, CBC-DDR-004	Done; choice awaits Amish
Requirements and traceability	Design lead	CBC-PRS-001, CBC-RTM-001	Done; 4 PRs at risk
Calculations	Design lead	CBC-CAL-001, `sizing.py`	Done (v0.4)
CAD at D3, every option	Design lead	`cad/src/model.py`, `cad/fidelity.json`, STEP, STL, drawings	Done; D3 pass
Build plan	Design lead	CBC-BLD-001	Done (paper only)
Regulatory, feasibility, risk, market	Design lead; regulatory consultant	CBC-REG-001, FEA-001, RSK-001, MKT-001	Done; many rows to assess
Renders and storefront images	Coordinator	`media/render-*.png`, `media/options.png`, cards	Rendered on Amish's Mac; options picture pending
Partner and co-design	Amish	A partner city with manual counts	Not started
Hardware build and tests	(on hold)	TRL 4 and later	On hold by the portfolio cap

Resources needed

Table 2. Resources. Effort is for the work after TRL 3 and is listed only as planned work.

Resource	Deliverable	Effort	Estimated cost (USD)
Partner city transport staff	Manual counts for comparison, site and pole access	to assess with the partner	to assess
Regulatory consultant	Rows marked to assess in CBC-REG-001	to assess	to assess
Design lead	Decisions O1 and O2, partner talks	a few days	none for the design
Parts for a first unit (TRL 4, on hold)	BOM lines 1 to 14	not started	274.00 (indicative prices)

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN exist	Passed
P1 Definition	URS, PRS, RSK, MKT exist; REG complete	Passed (REG rows marked to assess need a consultant)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR; STM and FEA closed	Reached except that STM has one open item (S8) and eight plausible rows, and the concept choice awaits Amish
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability complete	Reached; stage recorded in the kit 1.9.1 upgrade of 2026-10-31
P4 Verification	Test plans and lab test reports	On hold
P5 Validation	Field trials with a partner, certification status	On hold
P6 Commercialization	Launch plan, instructions for use, standard cost	On hold; the release gate needs `stem_check: pass`

Next steps and decision requested

1. Amish decides O2, the concept choice (A and B are within 10 %; A is the lean, with B as the hot-climate variant already agreed on 2026-09-25).
2. Amish decides O1: how to handle charging below -5 degrees C (limit the claim to cool sites now; ask FieldNode for a heated pocket or low-temperature cell).
3. The coordinator renders the product views and `media/options.png` on the Mac.
4. Close the plausible STEM rows (S3 to S5, S16 to S19, S24) by the tests that settle them once TRL 4 is allowed.

References

- CBC-PRB-001 v0.8, CBC-STM-001 v0.1, CBC-OPT-001 v0.1, CBC-URS-001 v0.1, CBC-PRS-001 v0.7, CBC-RTM-001, CBC-CAL-001 v0.4, CBC-BLD-001, CBC-REG-001, CBC-FEA-001, CBC-RSK-001, CBC-MKT-001, CBC-DEC-001, CBC-DDR-001 to 004 (this repo).
- Melexis MLX90640 datasheet, https://media.melexis.com/-/media/files/documents/datasheets/MLX90640-datasheet-melexis.pdf, accessed 2026-10-31.
- SparkFun IR Array Sensor 110 degree FOV MLX90640, https://www.sparkfun.com/ir-array-sensor-110-degree-fov-mlx90640.html, accessed 2026-10-31.
- Texas Instruments IWR6843, https://www.ti.com/product/IWR6843, and IWR6843ISK-ODS, https://www.ti.com/tool/IWR6843ISK-ODS, accessed 2026-10-31.
- NZ Transport Agency, cycle counting summary, https://nzta2.cwp.govt.nz/resources/sustainable-transport/cycle-counting-in-nz/summary, accessed 2026-10-31.
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- Department for Transport (England), Walking and cycling statistics 2023, https://www.gov.uk/government/statistics/walking-and-cycling-statistics-england-2023.
- World Health Organization, road safety news release of 13 December 2023, https://www.who.int/news/item/13-12-2023-despite-notable-progress-road-safety-remains-urgent-global-issue.